BKR MATRIZ — COMPLETE ARCHITECTURAL FLOWCHART

Full System Blueprint, Flowcharts, Workspaces & Relinking Specification

1. Executive Architectural Topology

BKR MATRIZ is a professional media preservation, intake management, cataloging, and backup system designed for high-fidelity preservation vaults, music labels, and recording archives. The architecture strictly enforces the isolation between **Physical Asset Storage**, **Project Files**, and **Database Authority**.

```
+----------------------------------------------------------------------------------------------------+
|                                    BKR MATRIZ ARCHITECTURAL LAYERS                                  |
+----------------------------------------------------------------------------------------------------+
| 1. PHYSICAL ASSET STORAGE (User Storages: HDs, SSDs, NAS, LTO, Removable Drives)                    |
|    - Holds original Master files (WAV, FLAC, RAW, MP4, PDF, etc.). Never modified or encapsulated.  |
+----------------------------------------------------------------------------------------------------+
| 2. PROJECT CONTAINERS (Two Distinct Project Types)                                                  |
|    - COLLECTION (.mtz): Single archival unit (Album, Session, Tape Batch, Preservation Collection).  |
|    - CATALOG (.bkm): High-level aggregator holding multiple Collections for Music Labels.           |
+----------------------------------------------------------------------------------------------------+
| 3. DATABASE & CACHE DATA LAYER (Independent of original media presence)                             |
|    - registro.sqlite: SOLE AUTHORITY. Items, Files, Checksums, Metadata, Provenance, Vault registrations.|
|    - indice.sqlite: Fast full-text search index, smart query views, and navigation cache.           |
|    - cache/ & Proxies: Waveform BLOBs (20 buckets/sec), thumbnails, technical metadata previews.     |
+----------------------------------------------------------------------------------------------------+
| 4. TWO-STAGE IN-MEMORY PERSISTENCE ENGINE                                                           |
|    - Live Mutations -> In-Memory State Map -> DIRTY Flag -> Explicit Save -> Atomically written to DB.  |
+----------------------------------------------------------------------------------------------------+
```

2. Global Lifecycle Flowchart: Launch to Exit

```
[ APPLICATION LAUNCH ]
       |
       v
[ START SCREEN (TELA INICIAL) ]
  ■■■ Mode A: Create New Collection (.mtz) -> Prompts Name & Path -> Initializes registro.sqlite -> Open Collecti
on
  ■■■ Mode B: Create New Catalog (.bkm)    -> Prompts Name & Path -> Initializes registro.sqlite -> Open Catalog
  ■■■ Action C: Open Collection (.mtz)    -> File Chooser -> Opens .mtz project container
  ■■■ Action D: Open Catalog (.bkm)       -> File Chooser -> Opens .bkm catalog container
  ■■■ Action E: Open Recent Files        -> Instant load of recent .mtz / .bkm project
       |
       v
[ PROJECT PRESENCE VERIFICATION (AssetRelinkEngine) ]
  ■■■ Master Assets Online (100% reachable) --------> Open Workspace normally
  ■■■ Master Assets Missing / Storage Disconnected -> Launch [ INITIAL RELINK MODAL ]
                                                       ■■■ [ LOCATE FILE ]  -> Pick 1 known file -> Root Inferenc
e -> Batch Relocate -> Open
                                                       ■■■ [ WORK OFFLINE ] -> Open immediately with cached wavef
orms & thumbnails
       |
       v
[ WORKSPACE NAVIGATION BAR & ACTIVE WORKSPACES ]
       |
       v
[ SAVE & CLOSE LIFECYCLE (Two-Stage Persistence) ]
  ■■■ File -> Save (Cmd+S) ----------> Writes in-memory relocations and metadata to registro.sqlite
  ■■■ File -> Save As (Cmd+Shift+S) -> Clones project container (.mtz / .bkm) to target directory
  ■■■ File -> Close / Quit -----------> If Dirty: Prompt [ SAVE ] [ DON'T SAVE ] [ CANCEL ]
                                       ■■■ Save: Commits in-memory state to disk -> Returns to Start Screen
                                       ■■■ Don't Save: Discards in-memory state (leaves disk unaltered) -> Return
s
                                       ■■■ Cancel: Aborts close action, stays in workspace
```

3. CATALOG Mode vs COLLECTION Mode Architecture

BKR MATRIZ operates in two distinct, non-interfering modes depending on the operational scope:

FEATURE / SCOPE	COLLECTION MODE (.mtz)	CATALOG MODE (.bkm)
Target Entity	Single preservation/archival collection (Reel, Cassette, Session, Film, Vinyl, Album).	Aggregator of multiple Collections designed for Music Labels and Publishers.
Contained Workspaces	Intake, Grid, List, Listening Station, Metadata, Tree, Duplicates, Analytics, Backup.	Catalog Hub, Inconsistencies Panel, Music Label Filtered Subsets, Child Collection Drill-Down.
Media Domain Scope	Full archival spectrum: 28+ physical and digital media types across all categories.	Music label relevant only: Releases, Songs, Stems, Music Videos, Artwork, Rights.
Collection Management	Direct file ingestion, metadata tagging, marker placement, physical preservation backup.	Links child .mtz collections, monitors collection status, relocates moved .mtz collections.
Navigation Hierarchy	Top-level work unit. Close returns to Start Screen (or parent Catalog if drilled down).	Top-level Catalog -> Drill-down to Child Collection -> Navigation Bar [Back to Catalog].

4. CATALOG MODE (.bkm) — Detailed Flowchart

```
[ OPEN CATALOG (.bkm) ]
       |
       v
+------------------------------------------------------------------------------------------------------------+
| CATALOG HUB WORKSPACE (CatalogHubComponent)                                                                |
|                                                                                                            |
|  [ + IMPORT COLLECTION... ]  [ LOCATE COLLECTION... ]     [ SEARCH / FILTER BAR: ALL | ONLINE | OFFLINE] |
|  ---------------------------------------------------------------------------------------------------------  |
|  STATUS   | COLLECTION NAME      | PATH / LOCATION              | TOTAL ASSETS | ACTIONS                   |
|  ---------+----------------------+------------------------------+--------------+-----------------------  |
|  ONLINE   | Album 1994 Remaster  | /Volumes/SSD/Album1994.mtz   | 24 items     | [ OPEN ] [ INFO ]         |
|  OFFLINE  | Tape Vault Batch B   | /Volumes/Archive/BatchB.mtz  | 142 items    | [ LOCATE... ] [ OPEN ] |
+------------------------------------------------------------------------------------------------------------+
       |                                                                        |
       | Double-Click Collection or [ OPEN ]                                    | Click [ LOCATE COLLECTION... ]
       v                                                                        v
[ LOAD CHILD COLLECTION (.mtz) ]                                         [ LOCATE COLLECTION FILE (.mtz) ]
  - Loads child Collection into full workspace                             - Validates .mtz container on disk
  - Top Bar displays: "Catalog Name  >  Collection Name"                   - Updates registro.sqlite collection lin
k
  - Top Bar displays button: [ < RETURN TO CATALOG ]                       - Marks Catalog project as DIRTY
  - Full Collection editing enabled                                        - Updates status badge to ONLINE
       |
       v
[ CLICK [ < RETURN TO CATALOG ] ]
  - Checks if Collection has unsaved changes (Two-stage prompt if dirty)
  - Closes child Collection and smoothly returns to Catalog Hub
```

5. COLLECTION Mode Workspaces & Functions

COLLECTION Mode provides 8 specialized workspaces designed for complete preservation and cataloging workflows.

5.1 INTAKE & QUARANTINE WORKSPACE (IntakeWorkspaceComponent)

Purpose: Secure landing zone for importing raw digital media files without altering originals.

```
[ DRAG & DROP FILES / FOLDERS or MENU "Add Files..." ]
       |
       v
[ MULTI-THREADED INGEST PIPELINE (ThreadPool & IngestQueue) ]
  ■■■ 1. Media Type & Category Auto-Classification (Audio, Video, Image, Document, 3D, Reel, Vinyl...)
  ■■■ 2. Automated Technical Metadata Extraction (Sample Rate, Bit Depth, Codec, Duration, Resolution, FPS)
  ■■■ 3. Embedded Metadata Extraction (EXIF, Dublin Core, ID3, RIFF/BWF, XMP, PDF Cataloging)
  ■■■ 4. Waveform & Metric Pre-computation (LUFS-I, True Peak, Sample Peak, LRA, Correlation, 20b/s Waveform)
  ■■■ 5. Cryptographic Checksum Generation (SHA-256 / MD5)
       |
       v
[ QUARANTINE REVIEW TABLE ]
  - Displays thumbnail, detected format, category, technical metrics, and OFFLINE badge if storage is missing.
  - Allows operator to adjust Category, Title, and Physical Medium before committing.
  - Actions: [ APPROVE & MOVE TO GRID ] | [ DISCARD ] | [ BATCH CATEGORIZE (Keys 1-9) ]
```

5.2 GRID & LIST WORKSPACE (MosaicoComponent)

Purpose: Main asset exploration interface supporting visual card tiles and structured tabular views.

```
[ GRID / LIST VIEW CONTROLS ]
  ■■■ View Mode Toggle: [ GRID VIEW ] (Card Tiles with Thumbnails) | [ LIST VIEW ] (Dense Table Columns)
  ■■■ Status Filter: [ ALL ] | [ ONLINE ] (Reachable on disk) | [ OFFLINE ] (Missing storage/drive)
  ■■■ Category Filter: Music, Video, Documents, Images, Artwork, Analog Masters, Multitracks...
  ■■■ Media Type Filter: Tape Reel, Cassette, Vinyl, DAT, CD, DVD, VHS, U-Matic, Betacam, Photo, Slide, PDF...
  ■■■ Search Bar: Instant full-text search across Title, Code, Artist, Medium, Notes, and Checksums.

[ ASSET INTERACTION ]
  ■■■ Single Click: Inspects asset in Metadata Panel & loads Waveform in Listening Station.
  ■■■ Double Click ONLINE Asset: Opens native system preview / default application.
  ■■■ Double Click OFFLINE Asset: Launches [ OFFLINE ASSET RELINK MODAL ] (Validation & Rebinding).
  ■■■ Drag Asset -> Vault Row: Plans incremental backup consolidation to target drive.
  ■■■ Context Menu: Rename (R), Remove from Backup (C), Delete, Custom Cover Art, Extract Stems.
```

5.3 METADATA INSPECTOR & PROVENANCE FORM (FichaTecnicaComponent)

Purpose: Comprehensive provenance recording, physical conservation grading, and technical archival data.

```
[ METADATA PANEL STRUCTURE ]
  ■■■ 1. Identification: Catalog Code (prefix-code), Title, Subtitle, Release Date, Artist, Producer.
  ■■■ 2. ORIGINAL SOURCE MEDIUM: Physical origin medium (Tape formulation, Vinyl cut, Master film gauge, etc.)
  ■■■ 3. Conservation & Physical Condition: State of preservation, physical defects, storage temperature/humidity
.
  ■■■ 4. Technical Digitization Details: ADC Converter, Transfer Speed, Equalization Curve, Tracking Alignment.
  ■■■ 5. Rights & Licensing: ISRC, ISWC, Rights Holder, Publishing, Reproduction Restrictions.
  ■■■ 6. Read-Only Original Metadata vs Project Metadata: Displays raw immutable file tags vs curated project dat
a.
```

5.4 LISTENING STATION & WAVEFORM (EstacaoEscutaComponent)

Purpose: Sub-100ms ultra-low latency audio auditioning and precision marker placement with offline support.

```
[ LISTENING STATION ARCHITECTURE ]
  ■■■ Waveform Rendering: Reads pre-calculated min/max buckets (20 buckets/sec) stored as BLOB in DB.
  ■■■ Offline Scrubbing: Displays exact waveform & loudness metrics EVEN WHEN STORAGE IS DISCONNECTED (< 1ms load
).
  ■■■ Transport Controls: Play/Pause (Space), Scrub, JKL shuttling, Volume, Loop. (Disabled when physical file of
fline).
  ■■■ Audio Metrics Footer: Real-time display of LUFS-I, Loudness Range (LRA), True Peak (dBTP), Phase Correlatio
n.
  ■■■ Archival Marker Management:
      ■■■ Range Markers (Dropout, Tape Flaw, Noise, Edit Region): Drag left/right handles to edit in/out times.
      ■■■ Instant Markers (Cue point, Track boundary, Annotation): Single-frame precision.
      ■■■ Non-destructive persistence: Markers saved in DB and optionally embedded into Broadcast WAV cue/iXML ch
unks.
```

5.5 TREE & HIERARCHY WORKSPACE (ArvoreExplorerComponent)

Purpose: Dual-view file tree reconciling Original Physical Folder Hierarchy with Logical Archival Tree.

```
[ DUAL TREE EXPLORER ]
  ■■■ Left Tree (Source / Origin Hierarchy): Preserves exact directory structure where files were ingested from.
  ■■■ Right Tree (Backup / Logical Hierarchy): Categorized folders for organized preservation packages.
```

5.6 DUPLICATES & ANALYTICS WORKSPACES

Duplicates Workspace: Scans SHA-256 hashes and audio fingerprint caches to detect exact byte copies and multi-version master takes.
Analytics Workspace: Displays real-time charts: Media Type distribution, Audio Resolution breakdown, Storage consumption per drive, and Archival Vulnerability audits.

5.7 BACKUP & CONSOLIDATION WORKSPACE (ConsolidacaoDialogo)

Purpose: Automated, verifiable multi-destination physical backups with non-destructive marker embedding.

```
[ BACKUP PIPELINE ]
  ■■■ 1. Drive Selection: Target external storage drive, LTO tape volume, or local mirror vault.
  ■■■ 2. Incremental Evaluation: Compares destination directory; skips unchanged files, calculates byte delta.
  ■■■ 3. Structured Mask Renaming: Formats output filenames: {prefix}_{code}_{title}_{sample_rate}.wav
  ■■■ 4. Marker Embedding (Optional): Non-destructively injects cue / iXML chunks into copied Broadcast WAV files
.
  ■■■ 5. Checksum Verification: Reads back copied bytes from destination and verifies SHA-256 against source.
  ■■■ 6. Package Manifest Generation: Writes manifest.txt, manifest.sqlite, and human-readable PDF Audit Report.
```

6. Asset Location, Offline State & Relinking Engine

The **AssetRelinkEngine** ensures seamless operation when projects are moved between machines, drives, or operating systems, enforcing strict two-stage persistence.

```
===========================================================================================================
                                ASSET RELINKING & OFFLINE ENGINE FLOWCHART
===========================================================================================================

       [ USER OPENS .mtz / .bkm PROJECT ]
                       |
                       v
       [ CHECK PHYSICAL FILE PRESENCE ]  <==========================================+
       (verificarPresencaAssets in registro.sqlite)                                 |
                       |                                                            |
            +----------+----------+                                                 |
            |                     |                                                 |
       All Present           Files Missing                                          |
            |                     |                                                 |
            v                     v                                                 |
     [ STATUS: ONLINE ]  [ LAUNCH INITIAL RELINK MODAL ]                            |
     (Project Ready)     (InitialRelinkDialog)                                      |
                                  |                                                 |
                       +----------+----------+                                      |
                       |                     |                                      |
                [ WORK OFFLINE ]      [ LOCATE FILE ]                               |
                       |                     |                                      |
                       v                     v                                      |
          [ OPEN IN OFFLINE MODE ]   [ USER PICKS 1 MATCHING FILE ]                 |
          - Waveforms available via DB       |                                      |
          - Thumbnails preserved in cache    v                                      |
          - Metadata fully editable  [ INFER NEW ROOT (inferirNovaRaiz) ]           |
          - OFFLINE badge on assets  - Matches path components tail backwards      |
          - Transport disabled       - Old Root: /Volumes/OldDrive/Batch1           |
                       |             - New Root: /Volumes/NewSSD/Batch1             |
                       |                     |                                      |
                       |                     v                                      |
                       |             [ BATCH RELOCATE (relocarColecaoEmMemoria) ]   |
                       |             - Validates all sibling candidate files        |
                       |             - Populates in-memory relocated paths map      |
                       |             - Marks Project as DIRTY                       |
                       |             - Assets turn ONLINE immediately               |
                       |                     |                                      |
                       +---------------------+--------------------------------------+
                                             |
                                             v
                      [ TWO-STAGE PERSISTENCE CYCLE (CRITICAL RULE) ]
                                             |
                      +----------------------+----------------------+
                      |                                             |
              [ USER PRESSES SAVE ]                       [ USER CLOSES PROJECT ]
              (Cmd+S / Menu Save)                                   |
                      |                                             v
                      v                              [ IS PROJECT DIRTY? ]
          [ ATOMIC DISK PERSISTENCE ]                               |
          - Updates registro.sqlite                  +--------------+--------------+
          - Persists new absolute paths              |                             |
          - Clears DIRTY flag                      DIRTY                       NOT DIRTY
          - Project remains clean                    |                             |
                                                     v                             v
                                      [ PROMPT UNSAVED CHANGES ]           [ CLOSE PROJECT ]
                                      [ SAVE ] [ DON'T SAVE ] [ CANCEL ]
                                         |           |            |
                      +-------------------+          |            +--> (Abort close)
                      |                              v
                      v                  [ DISCARD IN-MEMORY MAP ]
             [ SAVE TO DISK ]            - Leaves disk DB 100% untouched
             - Commits new paths         - Restores last saved state on reopen
             - Closes cleanly            - Closes cleanly
```

6.1 Individual Asset Relink Dialog (OfflineAssetRelinkDialog)

When an operator double-clicks an individual offline asset card, the modal inspects the single item: Expected Filename, Expected Size, Recorded SHA-256, and Last Known Storage. Upon selecting a new file, validarAsset() confirms the identity and updates the in-memory map for that asset only.

7. Complete Menu Bar & Keyboard Shortcut Map

MENU / SUBMENU	COMMAND	SHORTCUT	FUNCTION & BEHAVIOR
File ■ New	Collection (.mtz)...	—	Creates a new single preservation Collection project.
File ■ New	Catalog (.bkm)...	—	Creates a new Music Label Catalog aggregating collections.
File ■ Open	Collection (.mtz)...	—	Opens an existing .mtz collection project container.
File ■ Open	Catalog (.bkm)...	—	Opens an existing .bkm music catalog container.
File	Open Recent Files ■	—	Submenu listing recently accessed collections and catalogs.
File	Save Collection / Catalog	Cmd+S	Two-stage persistence: commits in-memory state to DB.
File	Save Collection / Catalog As...	Cmd+Shift+S	Clones project container to new destination directory.
File	Close Collection / Catalog	—	Prompts if dirty; cleanly unloads project to Start Screen.
File	Collection / Catalog Info...	—	Displays project statistics, database size, and vault paths.
File	Add Files...	—	Opens file picker to ingest raw files into Quarantine.
File	Quit	Cmd+Q	Exits application safely checking dirty project state.
Edit	Undo	Cmd+Z	Reverts last metadata edit or categorization change.
Edit	Rename Item(s)...	R	Batch or single title renaming for mask generation.
Edit	Remove Selected from Backup	C	Excludes item from backup plan (retains in project).
Navigation	Play / Pause Audio	Space	Toggles transport in Listening Station.
Navigation	Batch Categorize 1-9	1 .. 9	Instantly applies media category to selected batch.
Preferences	Theme / AI Key / Device	—	Configures audio interface, AI vision key, dark/light theme.

8. Provenance Audit Model & Master Immutability

Immutable Master Rule (P1): Original files registered with eh_master = 1 can never be overwritten or modified. Any processing produces a registered derivative record referencing the master ID.
Audit Trail: Every metadata edit, category change, relocation event, marker adjustment, and backup consolidation is logged to the proveniencia table with ISO-8601 timestamp, operator signature, and previous/new checksum hashes.